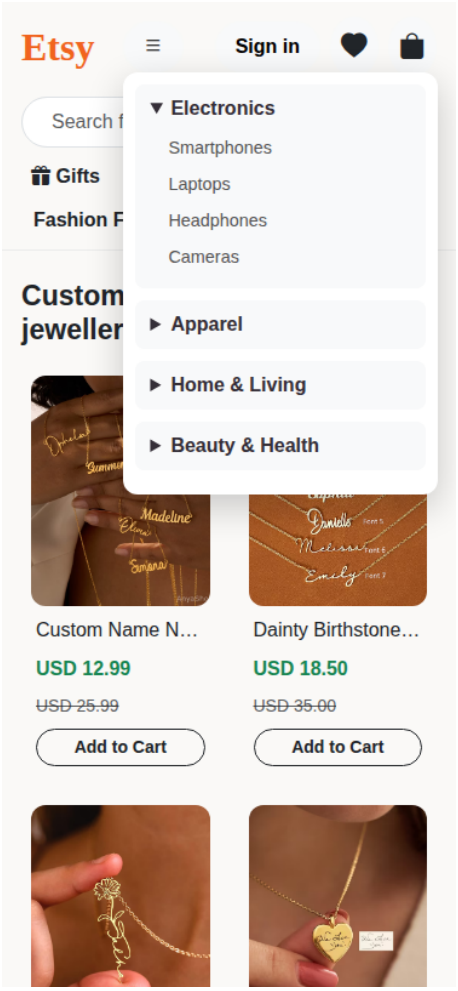
M.Hashim — FA24-BCS-119

Web Technologies Fundamentals — Midterm (Etsy clone)
BCS(B) | N = 119 | BP = 991 px

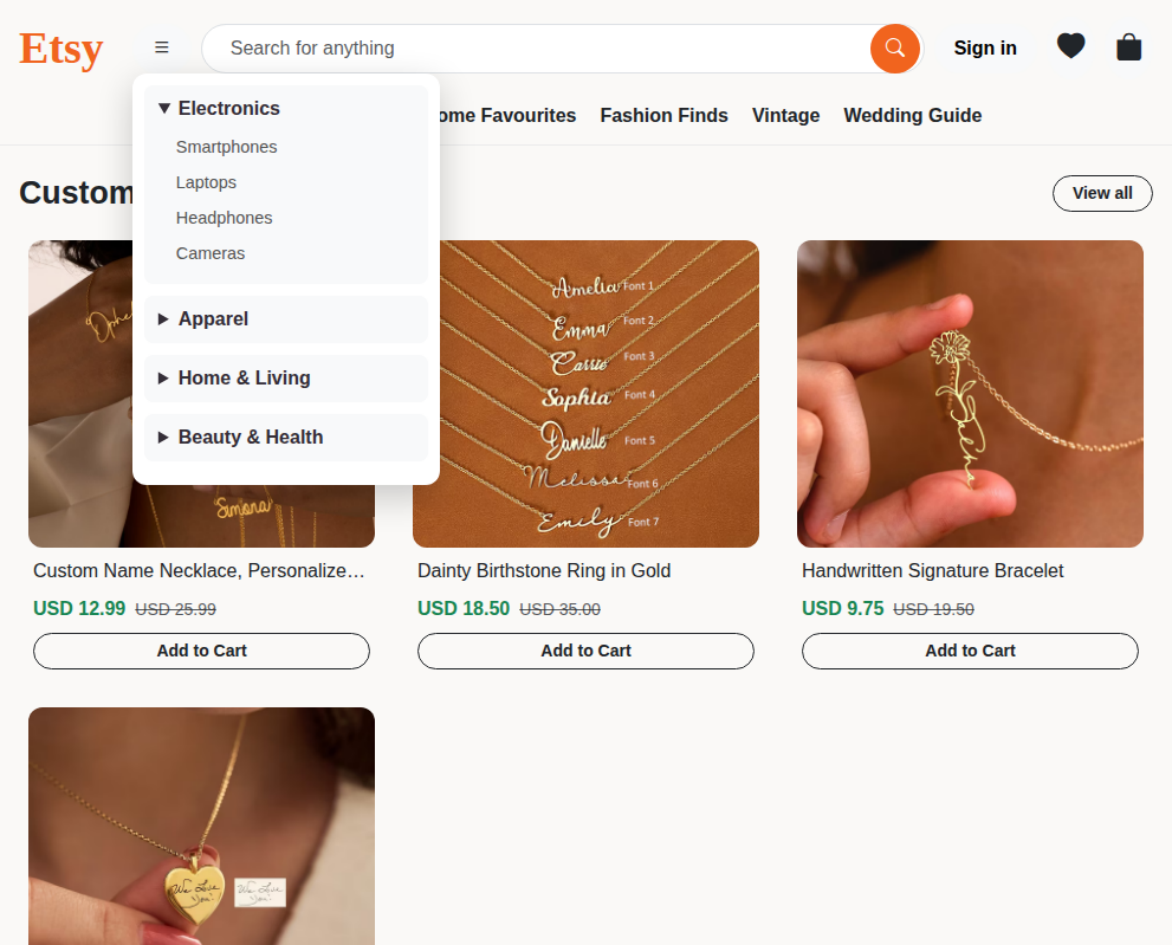
Screenshots (375 px, 991 px, 1280 px)

Task 1 — Shop by Category menu

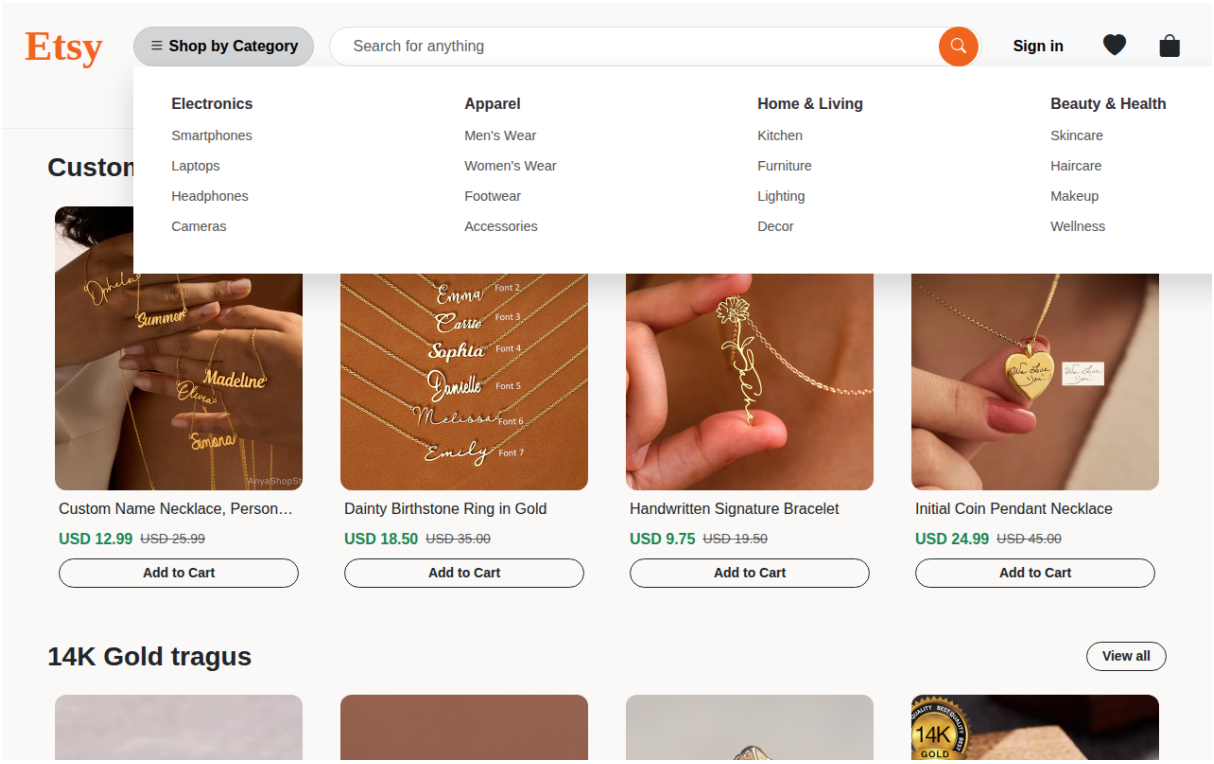
Task 1 — 375 px: accordion opens under the category button



Task 1 — 991 px (BP): accordion menu

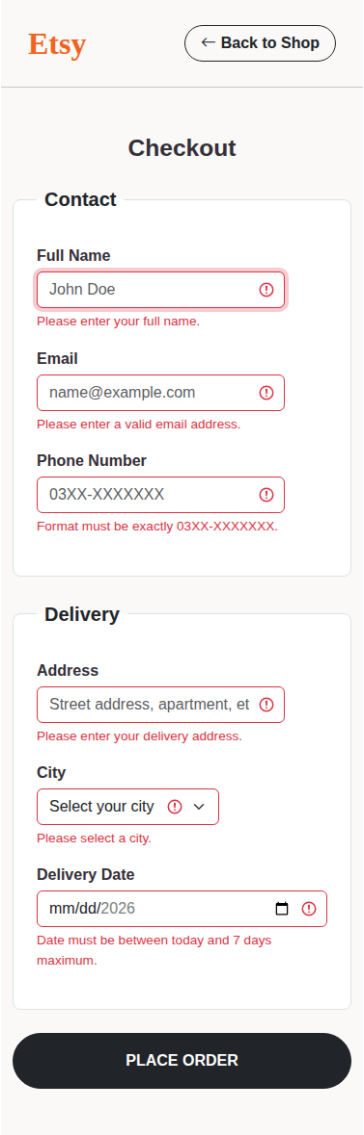


Task 1 — 1280 px: full-width mega menu on hover

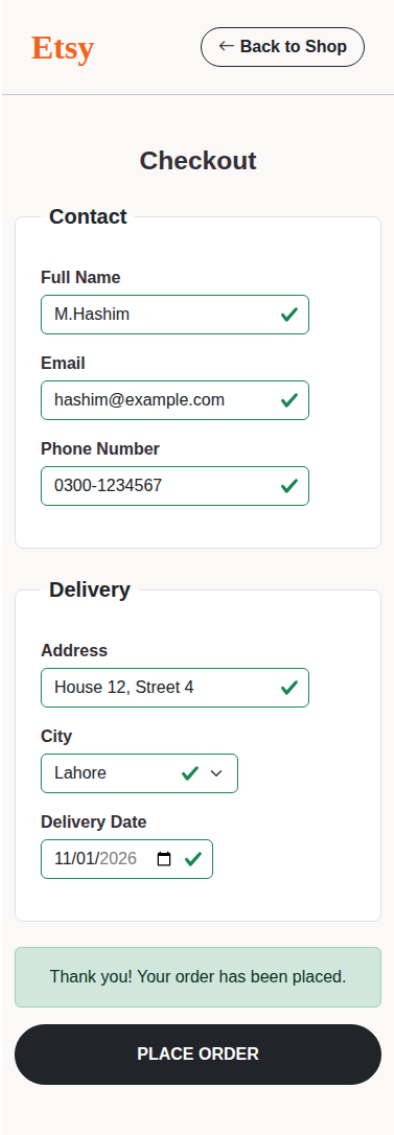


Task 2 — Checkout form validation

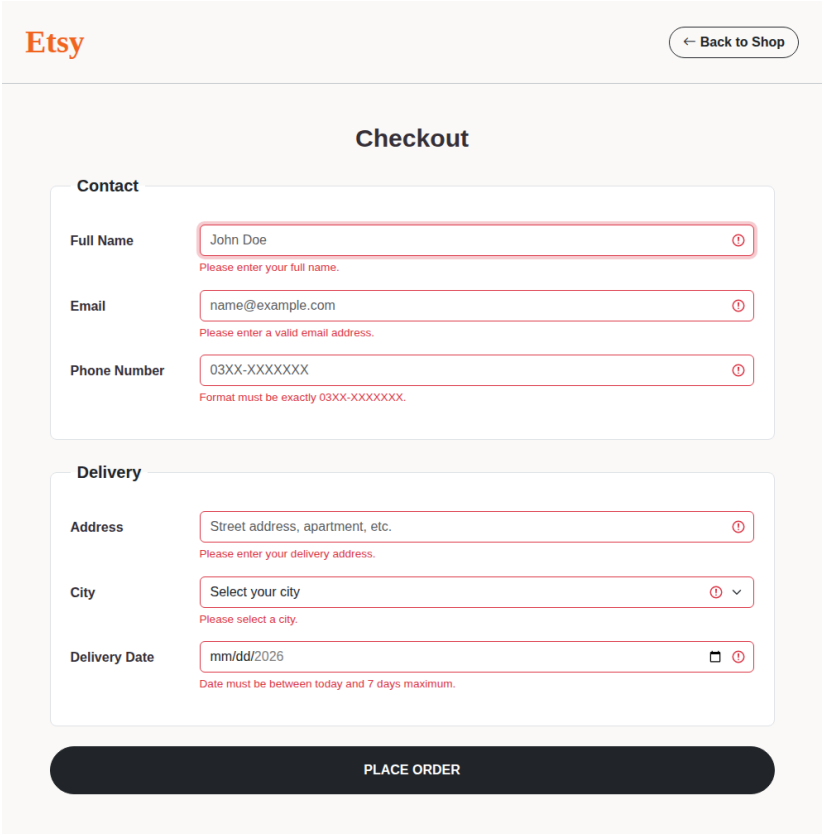
Task 2 — 375 px: empty submit (errors)



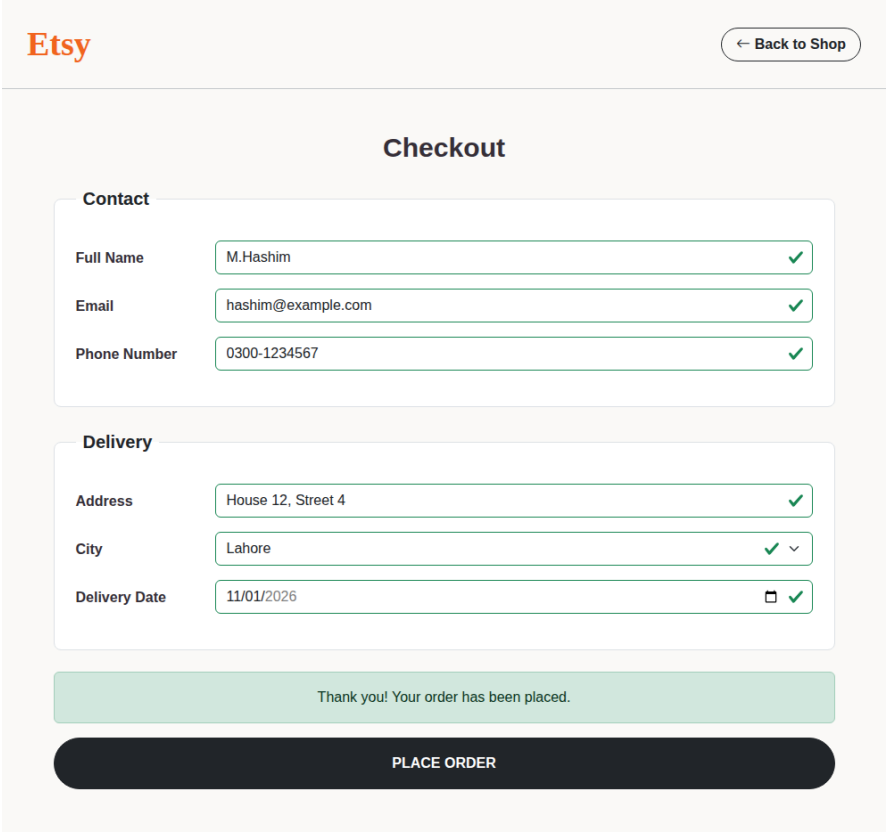
Task 2 — 375 px: valid submit



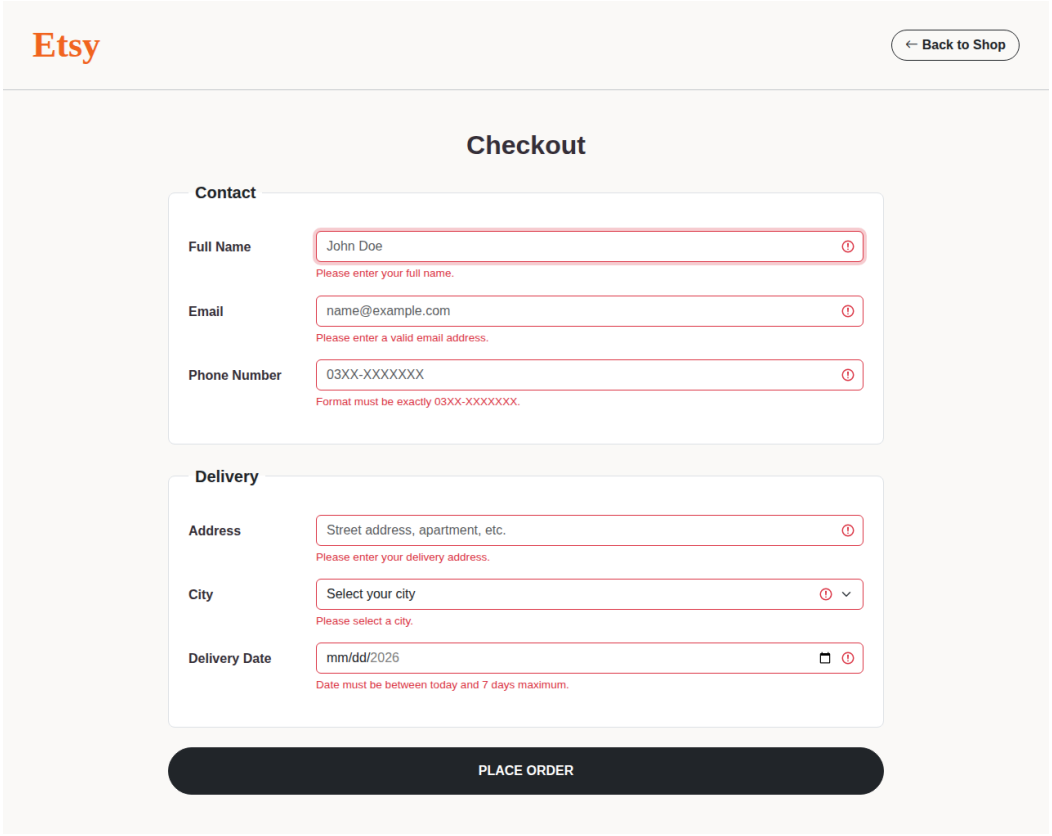
Task 2 — 991 px (BP): empty submit shows errors



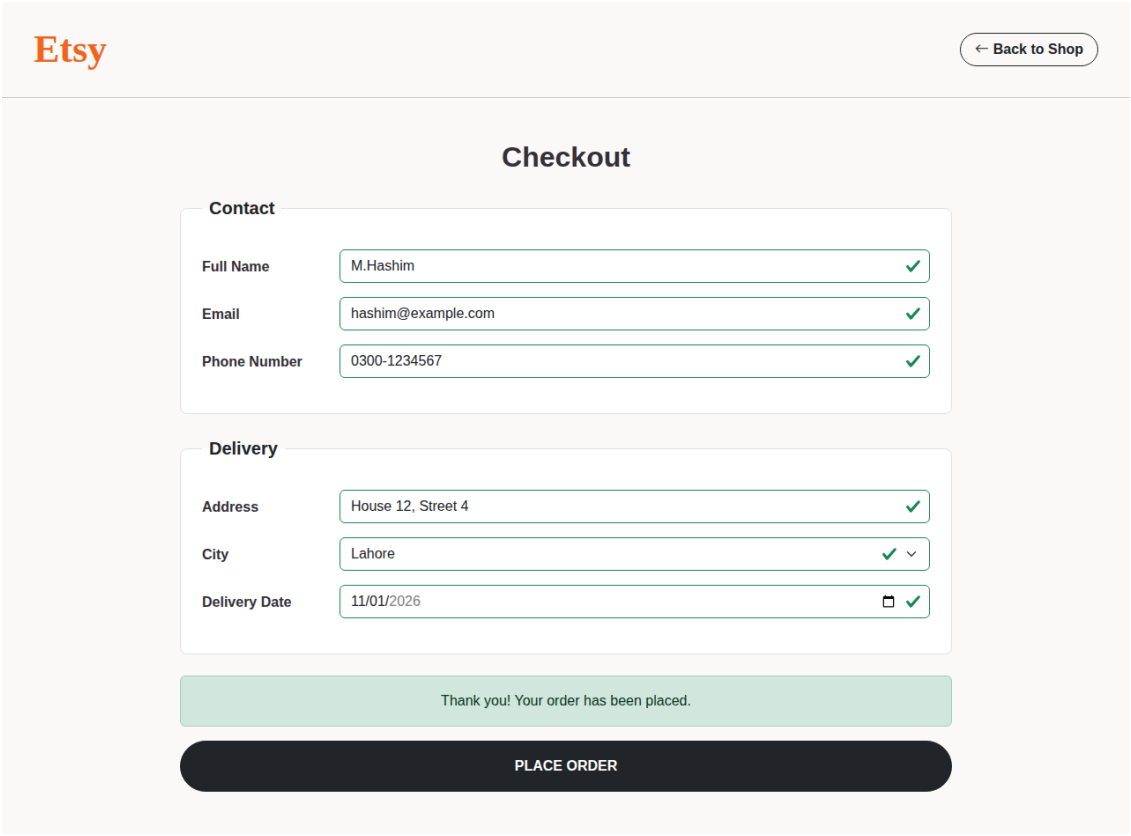
Task 2 — 991 px (BP): valid submit



Task 2 — 1280 px: empty submit shows errors

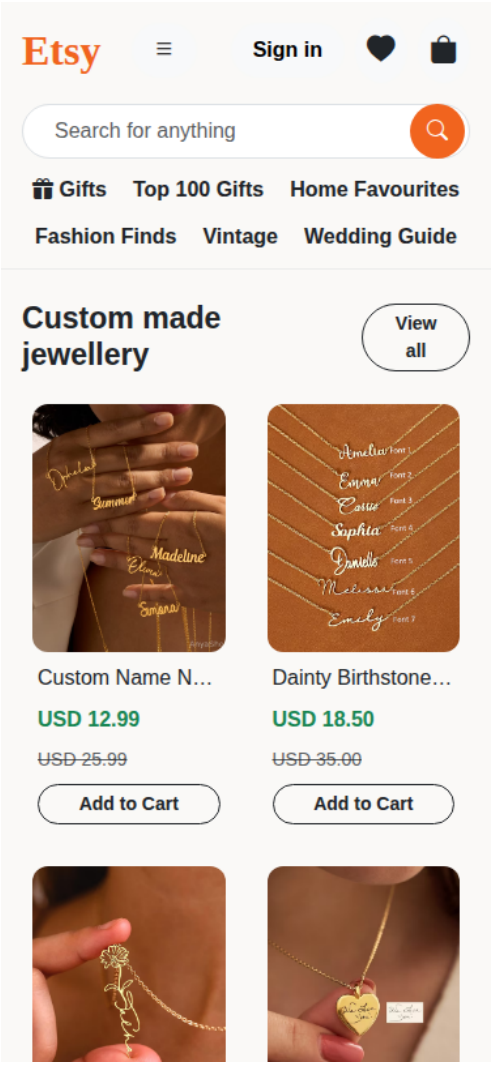


Task 2 — 1280 px: valid submit

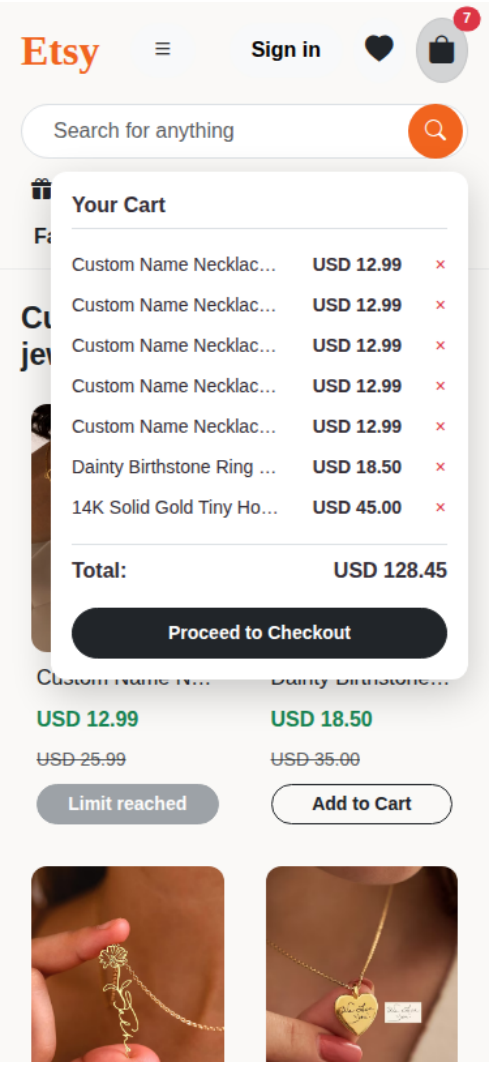


Task 3 — Shopping cart

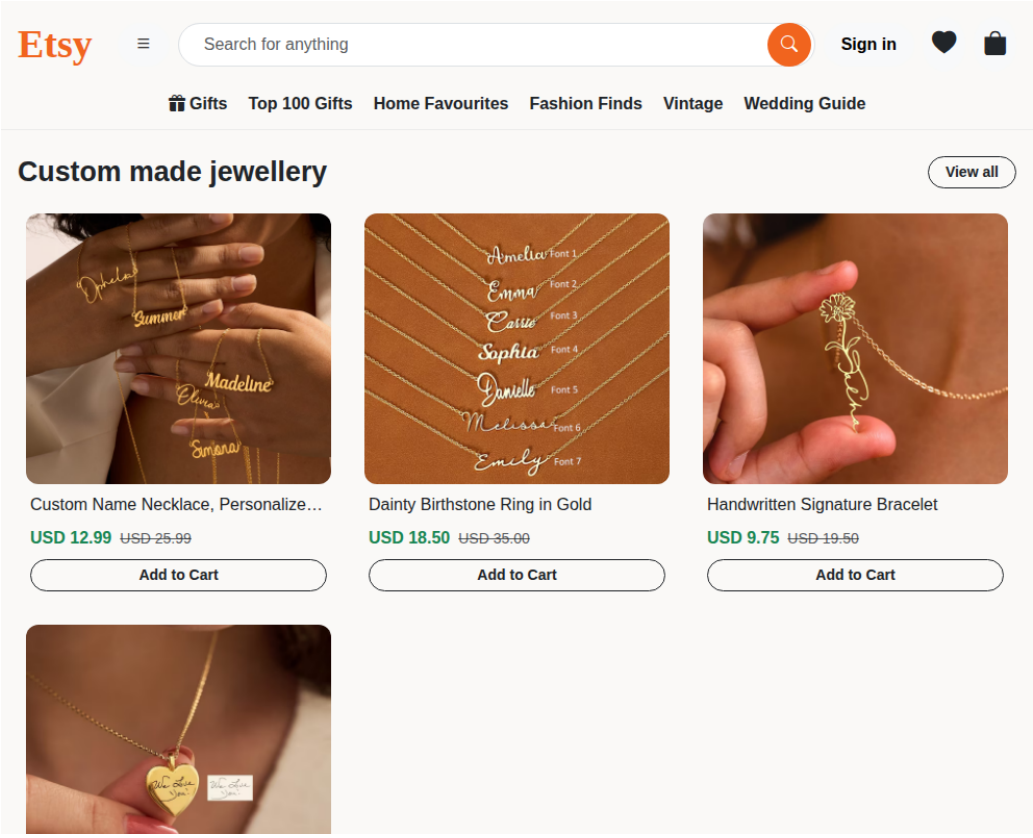
Task 3 — 375 px: before (Add to Cart buttons)



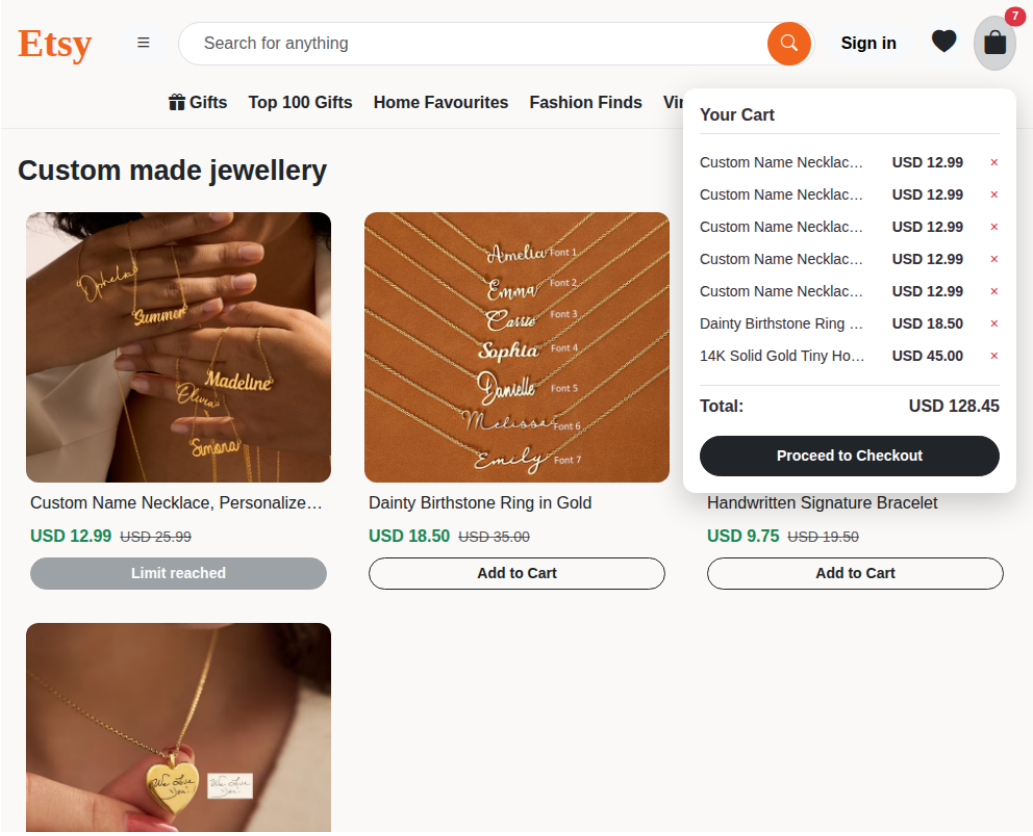
Task 3 — 375 px: after adding items (limit reached, mini-cart open)



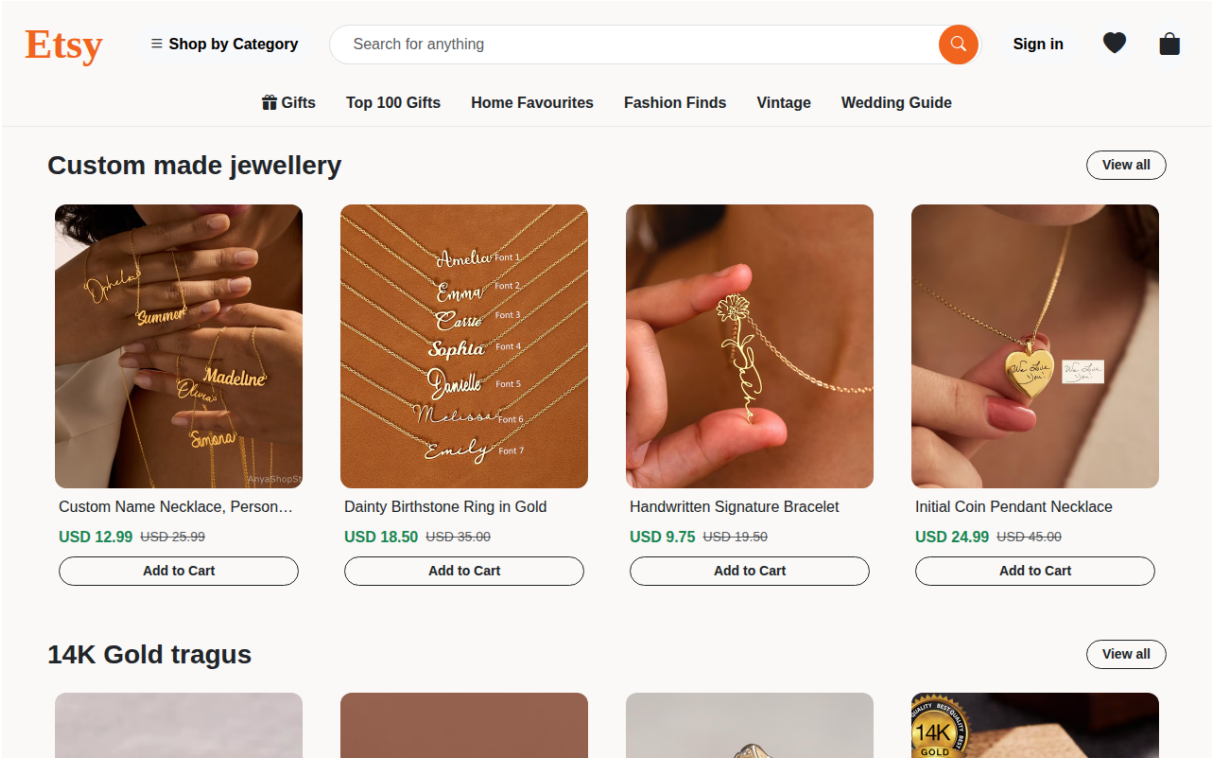
Task 3 — 991 px (BP): before



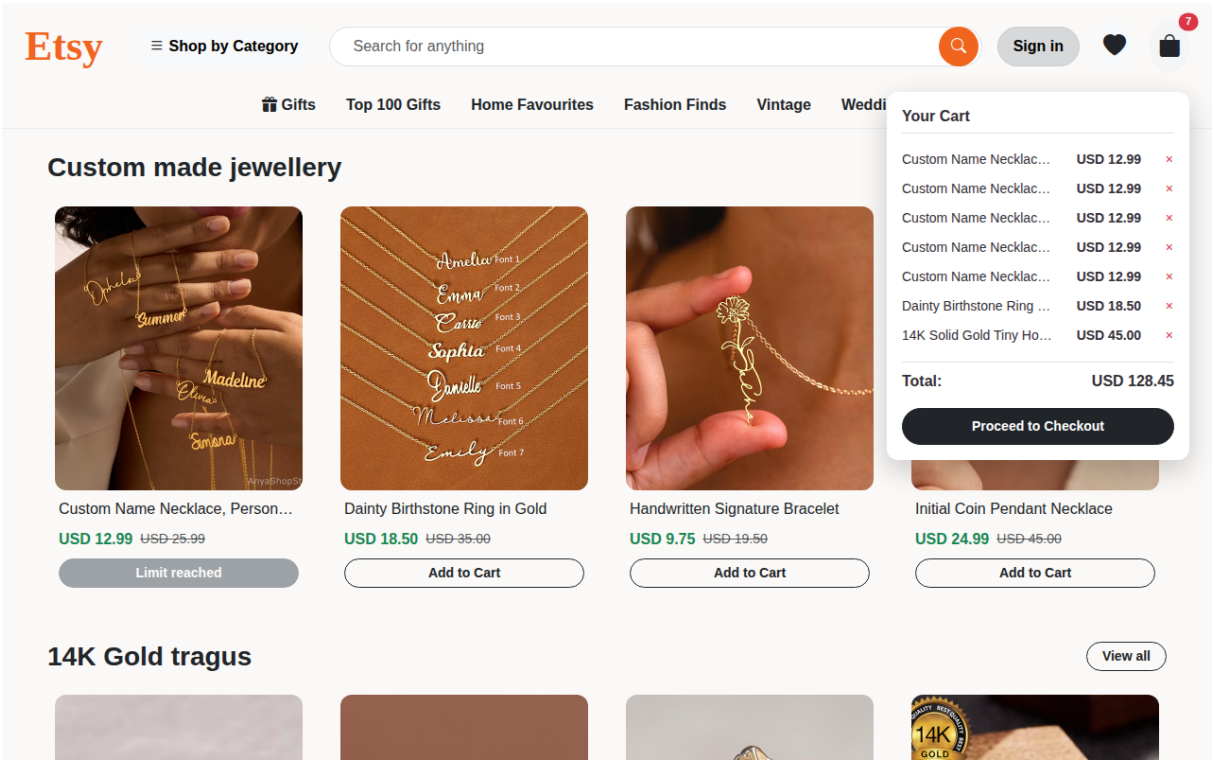
Task 3 — 991 px (BP): after using the cart



Task 3 — 1280 px: before



Task 3 — 1280 px: after using the cart



Code added or changed

Task 1 — Shop by Category menu

index.html — menu markup (desktop mega-panel + mobile accordion)

```
<!-- SHOP BY CATEGORY WRAPPER -->
            <ul class="navbar-nav me-auto mb-0 list-unstyled">
                <li class="nav-item dropdown-parent position-relative">
                    <button class="btn btn-light rounded-pill fw-semibold text-nowrap px-3 py-2"
                        type="button">
                        <i class="bi bi-list"></i>
                        <span class="d-none d-lg-inline">Shop by Category</span>
                    </button>

                    <!-- 1. DESKTOP FULL-WIDTH MEGA-PANEL -->
                    <div class="mega-menu-panel">
                        <div class="mega-menu-container">
                            <div class="mega-col">
                                <h3>Electronics</h3>
                                <ul class="list-unstyled">
                                    <li><a href="#">Smartphones</a></li>
                                    <li><a href="#">Laptops</a></li>
                                    <li><a href="#">Headphones</a></li>
                                    <li><a href="#">Cameras</a></li>
                                </ul>
                            </div>
                            <div class="mega-col">
                                <h3>Apparel</h3>
                                <ul class="list-unstyled">
                                    <li><a href="#">Men's Wear</a></li>
                                    <li><a href="#">Women's Wear</a></li>
                                    <li><a href="#">Footwear</a></li>
                                    <li><a href="#">Accessories</a></li>
                                </ul>
                            </div>
                            <div class="mega-col">
                                <h3>Home & Living</h3>
                                <ul class="list-unstyled">
                                    <li><a href="#">Kitchen</a></li>
                                    <li><a href="#">Furniture</a></li>
                                    <li><a href="#">Lighting</a></li>
                                    <li><a href="#">Decor</a></li>
                                </ul>
                            </div>
                            <div class="mega-col">
                                <h3>Beauty & Health</h3>
                                <ul class="list-unstyled">
                                    <li><a href="#">Skincare</a></li>
                                    <li><a href="#">Haircare</a></li>
                                    <li><a href="#">Makeup</a></li>
                                    <li><a href="#">Wellness</a></li>
                                </ul>
                            </div>
                        </div>
                    </div>

                    <!-- 2. MOBILE ACCORDION -->
                    <div class="mobile-accordion-group">
                        <details>
                            <summary>Electronics</summary>
                            <ul class="list-unstyled">
                                <li><a href="#">Smartphones</a></li>
                                <li><a href="#">Laptops</a></li>
                                <li><a href="#">Headphones</a></li>
                                <li><a href="#">Cameras</a></li>
                            </ul>
                        </details>
                        <details>
                            <summary>Apparel</summary>
                            <ul class="list-unstyled">
                                <li><a href="#">Men's Wear</a></li>
                                <li><a href="#">Women's Wear</a></li>
                                <li><a href="#">Footwear</a></li>
                                <li><a href="#">Accessories</a></li>
                            </ul>
                        </details>
                        <details>
                            <summary>Home & Living</summary>
                            <ul class="list-unstyled">
                                <li><a href="#">Kitchen</a></li>
                                <li><a href="#">Furniture</a></li>
```

```
                            <li><a href="#">Lighting</a></li>
                            <li><a href="#">Decor</a></li>
                        </ul>
                    </details>
                    <details>
                        <summary>Beauty & Health</summary>
                        <ul class="list-unstyled">
                            <li><a href="#">Skincare</a></li>
                            <li><a href="#">Haircare</a></li>
                            <li><a href="#">Makeup</a></li>
                            <li><a href="#">Wellness</a></li>
                        </ul>
                    </details>
                </div>
            </li>
        </ul>
```

style.css — mega menu, accordion and responsive rules (mobile accordion changed to a dropdown under the button at max-width 991px)

```
/* --- SHOP BY CATEGORY MEGA-MENU & MOBILE ACCORDION STYLES --- */

.mobile-accordion-group {
    display: none;
}

.dropdown-parent {
    position: relative;
}

/* Desktop Full-Width Mega Panel */
.mega-menu-panel {
    position: absolute;
    top: 100%;
    left: 0;
    width: 100vw;
    background: #ffffff;
    box-shadow: 0 15px 30px rgba(0, 0, 0, 0.15);
    opacity: 0;
    visibility: hidden;
    transition: opacity 0.2s ease, visibility 0.2s ease;
    z-index: 9999;
}

.mega-menu-container {
    max-width: 1300px;
    margin: 0 auto;
    display: flex;
    justify-content: space-between;
    gap: 40px;
    padding: 30px 40px;
}

.mega-col {
    flex: 1;
}

.mega-col h3 {
    font-size: 1rem;
    font-weight: bold;
    margin-bottom: 12px;
    color: #352f38;
}

.mega-col ul {
    list-style: none;
    padding: 0;
    margin: 0;
}

.mega-col li {
    margin-bottom: 8px;
}

.mega-col a {
    text-decoration: none;
    color: #555;
    font-size: 0.9rem;
}

.mega-col a:hover {
    color: #000;
```

```
    text-decoration: underline;
}

.dropdown-parent:hover .mega-menu-panel,
.dropdown-parent:focus-within .mega-menu-panel {
    opacity: 1;
    visibility: visible;
}


/* --- RESPONSIVE BREAKPOINTS --- */

@media (max-width: 991px) {
    .logo {
        font-size: 38px;
    }
    .product-img {
        height: 260px;
    }
    .gift-img {
        height: 300px;
    }
    .footer-mission {
        font-size: 34px;
    }

    .mega-menu-panel {
        display: none !important;
    }

    .mobile-accordion-group {
        position: absolute;
        top: 100%;
        left: 0;
        width: 260px;
        background: #ffffff;
        padding: 10px;
        border-radius: 12px;
        box-shadow: 0 10px 30px rgba(0, 0, 0, 0.2);
        z-index: 9999;
    }

    .dropdown-parent:hover .mobile-accordion-group,
    .dropdown-parent:focus-within .mobile-accordion-group {
        display: block;
    }

    .mobile-accordion-group details {
        margin-bottom: 10px;
        background: #f8f9fa;
        padding: 8px 12px;
        border-radius: 8px;
    }

    .mobile-accordion-group summary {
        font-weight: 600;
        cursor: pointer;
        color: #352f38;
    }

    .mobile-accordion-group ul {
        list-style: none;
        padding-left: 15px;
        margin-top: 8px;
    }

    .mobile-accordion-group li {
        margin-bottom: 6px;
    }

    .mobile-accordion-group a {
        text-decoration: none;
        color: #555;
        font-size: 0.9rem;
    }
}

@media (max-width: 575px) {
    .logo {
        font-size: 32px;
    }
    .product-img {
        height: 190px;
```

```
    }
    .gift-img {
        height: 220px;
    }
    .footer-mission {
        font-size: 27px;
    }
}
```

Task 2 — Checkout form validation

checkout.html — added success message above the button, and the script tag before </body>

```
        <div id="orderMessage" class="alert alert-success text-center d-none" role="alert">Thank you!
            Your order has been placed.</div>

        <button type="submit" class="btn btn-dark w-100 py-3 fw-bold rounded-pill text-uppercase">Place
            Order</button>

    ...
<script src="checkout.js"></script>
</body>
```

checkout.js (new file)

```
document.addEventListener('DOMContentLoaded', () => {
    const form = document.querySelector('.checkout-form');
    const message = document.getElementById('orderMessage');
    const dateInput = document.getElementById('deliveryDate');
    const fields = form.querySelectorAll('input, select');

    // Delivery date must be between today and 7 days from today
    const toISO = (d) => {
        const m = String(d.getMonth() + 1).padStart(2, '0');
        const day = String(d.getDate()).padStart(2, '0');
        return `${d.getFullYear()}-${m}-${day}`;
    };
    const today = new Date();
    const maxDay = new Date();
    maxDay.setDate(today.getDate() + 7);
    dateInput.min = toISO(today);
    dateInput.max = toISO(maxDay);

    // Check one field using the browser's built-in rules (required, type, pattern, min, max)
    function checkField(field) {
        const ok = field.checkValidity();
        field.classList.toggle('is-invalid', !ok);
        field.classList.toggle('is-valid', ok);
        return ok;
    }

    fields.forEach((field) => {
        field.addEventListener('blur', () => checkField(field));
        field.addEventListener('input', () => {
            if (field.classList.contains('is-invalid')) checkField(field);
        });
        field.addEventListener('change', () => checkField(field));
    });

    form.addEventListener('submit', (e) => {
        e.preventDefault();
        let allValid = true;

        fields.forEach((field) => {
            if (!checkField(field)) allValid = false;
        });

        if (allValid) {
            message.classList.remove('d-none');
        } else {
            message.classList.add('d-none');
            form.querySelector('.is-invalid').focus();
        }
    });
});
```

checkout.css (new file; was linked in checkout.html but missing)

```
/* Checkout form layout: label on the left, input on the right */
.form-row {
    display: flex;
    align-items: flex-start;
    gap: 16px;
}

.form-row label {
    flex: 0 0 140px;
    padding-top: 8px;
    font-weight: 600;
```

```
}

.input-wrapper {
    flex: 1;
}

/* Error messages stay hidden until the field is invalid */
.error-message {
    display: none;
    color: #dc3545;
    font-size: 0.85rem;
    margin-top: 4px;
}

.is-invalid ~ .error-message {
    display: block;
}

@media (max-width: 575px) {
    .form-row {
        flex-direction: column;
        gap: 4px;
    }

    .form-row label {
        flex: none;
        padding-top: 0;
    }
}
```

Task 3 — Shopping cart

index.html — script tag added before </body> (script.js was never loaded)

```
<script src="script.js"></script>
</body>
```

script.js — MAX_LIMIT changed from 3 to 5, and e.preventDefault() added in the Add to Cart click handler

```
document.addEventListener('DOMContentLoaded', () => {
    let cart = [];
    const MAX_LIMIT = 5; // (N mod 4) + 2 where N = 119 -> (119 % 4) + 2 = 3 + 2 = 5

    // Locate the cart icon in the navbar
    const cartLink = document.querySelector('a[href="checkout.html"]');
    if (!cartLink) return;

    const parentDiv = cartLink.parentElement;
    parentDiv.style.position = 'relative';

    // Create and attach Badge element
    const badge = document.createElement('span');
    badge.className = 'position-absolute top-0 start-100 translate-middle badge rounded-pill bg-danger
        cart-badge';
    badge.style.display = 'none';
    cartLink.appendChild(badge);

    // Create and attach Mini-Cart Panel element
    const miniCart = document.createElement('div');
    miniCart.className = 'mini-cart-panel';
    miniCart.innerHTML = `
        <h6 class="fw-bold mb-3 border-bottom pb-2">Your Cart</h6>
        <div class="mini-cart-items mb-3"></div>
        <div class="d-flex justify-content-between fw-bold mb-3 border-top pt-2">
            <span>Total:</span>
            <span class="cart-total">USD 0.00</span>
        </div>
        <a href="checkout.html" class="btn btn-dark w-100 rounded-pill py-2 btn-sm fw-semibold">Proceed
            to Checkout</a>
    `;
    parentDiv.appendChild(miniCart);

    // Toggle Mini-Cart panel on icon click
    cartLink.addEventListener('click', (e) => {
        e.preventDefault();
        miniCart.classList.toggle('show');
    });

    // Close mini-cart when clicking outside
    document.addEventListener('click', (e) => {
        if (!parentDiv.contains(e.target)) {
            miniCart.classList.remove('show');
        }
    });

    // Inject "Add to Cart" buttons into each product card dynamically
    const productCards = document.querySelectorAll('.product-card');
    productCards.forEach((card) => {
        const body = card.querySelector('.card-body') || card;

        const titleEl = card.querySelector('.card-title') || card.querySelector('p');
        const priceEl = card.querySelector('.sale-price');

        if (!titleEl || !priceEl) return;

        const productName = titleEl.textContent.trim();
        const priceValue = parseFloat(priceEl.textContent.replace('USD', '').trim());

        const addBtn = document.createElement('button');
        addBtn.className = 'btn btn-outline-dark btn-sm rounded-pill w-100 mt-2 fw-semibold add-to-cart-
            btn';
        addBtn.type = 'button';
        addBtn.textContent = 'Add to Cart';
        body.appendChild(addBtn);

        addBtn.addEventListener('click', (e) => {
            e.preventDefault(); // button sits inside the product <a> link on some cards
            const currentCount = cart.filter(item => item.name === productName).length;

            if (currentCount < MAX_LIMIT) {
```

```
                cart.push({ name: productName, price: priceValue });
                updateCartUI();
            }

            const updatedCount = cart.filter(item => item.name === productName).length;
            if (updatedCount >= MAX_LIMIT) {
                addBtn.textContent = 'Limit reached';
                addBtn.disabled = true;
                addBtn.classList.remove('btn-outline-dark');
                addBtn.classList.add('btn-secondary');
            }
        });
    });

    // Update Badge, List, Totals using reduce(), and Button States
    function updateCartUI() {
        if (cart.length > 0) {
            badge.style.display = 'inline-block';
            badge.textContent = cart.length;
        } else {
            badge.style.display = 'none';
        }

        const itemsContainer = miniCart.querySelector('.mini-cart-items');
        itemsContainer.innerHTML = '';

        if (cart.length === 0) {
            itemsContainer.innerHTML = '<p class="text-muted small text-center mb-0">Your cart is
                empty.</p>';
        } else {
            cart.forEach((item, index) => {
                const row = document.createElement('div');
                row.className = 'd-flex justify-content-between align-items-center mb-2 small';
                row.innerHTML = `
                    <span class="text-truncate me-2" style="max-width: 160px;">${item.name}</span>
                    <span class="fw-semibold text-nowrap">USD ${item.price.toFixed(2)}</span>
                    <button class="btn btn-sm text-danger p-0 ms-2 remove-item" data-index="${index}"
                        title="Remove">&times;</button>
                `;
                itemsContainer.appendChild(row);
            });
        }

        // Calculate total price using reduce()
        const total = cart.reduce((sum, item) => sum + item.price, 0);
        miniCart.querySelector('.cart-total').textContent = `USD ${total.toFixed(2)}`;

        // Attach Remove Handlers
        document.querySelectorAll('.remove-item').forEach(btn => {
            btn.addEventListener('click', (e) => {
                const index = parseInt(e.target.getAttribute('data-index'));
                const removedItemName = cart[index].name;

                cart.splice(index, 1);
                updateCartUI();

                // Re-enable corresponding product button if below limit
                productCards.forEach(card => {
                    const titleEl = card.querySelector('.card-title') || card.querySelector('p');
                    if (titleEl && titleEl.textContent.trim() === removedItemName) {
                        const addBtn = card.querySelector('.add-to-cart-btn');
                        const currentCount = cart.filter(item => item.name === removedItemName).length;
                        if (addBtn && currentCount < MAX_LIMIT) {
                            addBtn.textContent = 'Add to Cart';
                            addBtn.disabled = false;
                            addBtn.classList.remove('btn-secondary');
                            addBtn.classList.add('btn-outline-dark');
                        }
                    }
                });
            });
        });
    }

    updateCartUI();
});
```

style.css — mini-cart panel, cart badge and small-screen header fix (added)

```
/* --- CART BADGE & MINI-CART PANEL (Task 3) --- */

a[aria-label="Checkout"] {
    position: relative;
```

```
}

.cart-badge {
    font-size: 0.7rem;
}

.mini-cart-panel {
    display: none;
    position: absolute;
    top: calc(100% + 10px);
    right: 0;
    width: 320px;
    max-width: calc(100vw - 24px);
    background: #ffffff;
    border-radius: 12px;
    padding: 16px;
    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.2);
    z-index: 10000;
}

.mini-cart-panel.show {
    display: block;
}


/* --- HEADER FIX: stop navbar overflowing on small screens --- */

header form[role="search"] {
    min-width: 0;
}

@media (max-width: 575px) {
    header .navbar > .d-flex {
        flex-wrap: wrap;
    }

    header form[role="search"] {
        order: 10;
        flex: 0 0 100%;
        margin-top: 4px;
    }
}
```